

Hand-in-the-Loop: Improving Dexterous VLA via Seamless Interventional Correction

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Abstract

Vision-Language-Action (VLA) models suffers from the accumulation of compounding errors when applied to dexterous manipulation because of high-degree-of-freedom (DoF) action spaces. While Interactive Imitation Learning (IIL) has proven effective for policy refinement through human takeover data, its deployment in dexterous manipulation remains challenging due to “gesture jumps” at the moment of intervention. We present Hand-in-the-Loop (HandITL), a seamless interventional method that fuses human correction and policy rollout without bringing gesture jump at takeover moment for bimanual dexterous system. By seamlessly fusing human corrective intent with the model’s ongoing actions, HandITL reduces the intervention jitter by 99.8% and preserves robust manipulation capabilities post-takeover with improving mean finish time by 19.1% and reducing grasp failures by 87.5%, compared to teleoperation. Our method is designed to support complex bimanual coordination, tool use, and fine-grained long-horizon tasks. Extensive real-world experiments on dexterous VLA models demonstrate that HandITL significantly outperforms the policy trained with standard teleoperation data by 19% on average on three long-horizon dexterous tasks, effectively mitigating compounding errors and substantially boosting task scores.

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Project Page: [xxx](#)

1 Introduction

Vision-Language-Action (VLA) models have achieved remarkable success in robotic manipulation [12, 34]. However, extending these capabilities from parallel-jaw grippers to high-degree-of-freedom (DoF) anthropomorphic hands remains a formidable challenge. Consequently, deploying VLA policies on real dexterous systems often suffers from post-deployment distribution shift. During contact-rich, long-horizon execution, minor deviations in fingertip pose rapidly compound, pushing the robot into out-of-distribution (OOD) states where the base policy fails catastrophically [21, 25]. To mitigate this problem, Interactive Imitation Learning (IIL), such as Dataset Aggregation (DAgger) [11, 21], allows human operators to intervene during rollouts. This explicitly teaches the robot recovery behaviors, ensuring ongoing task success while simultaneously

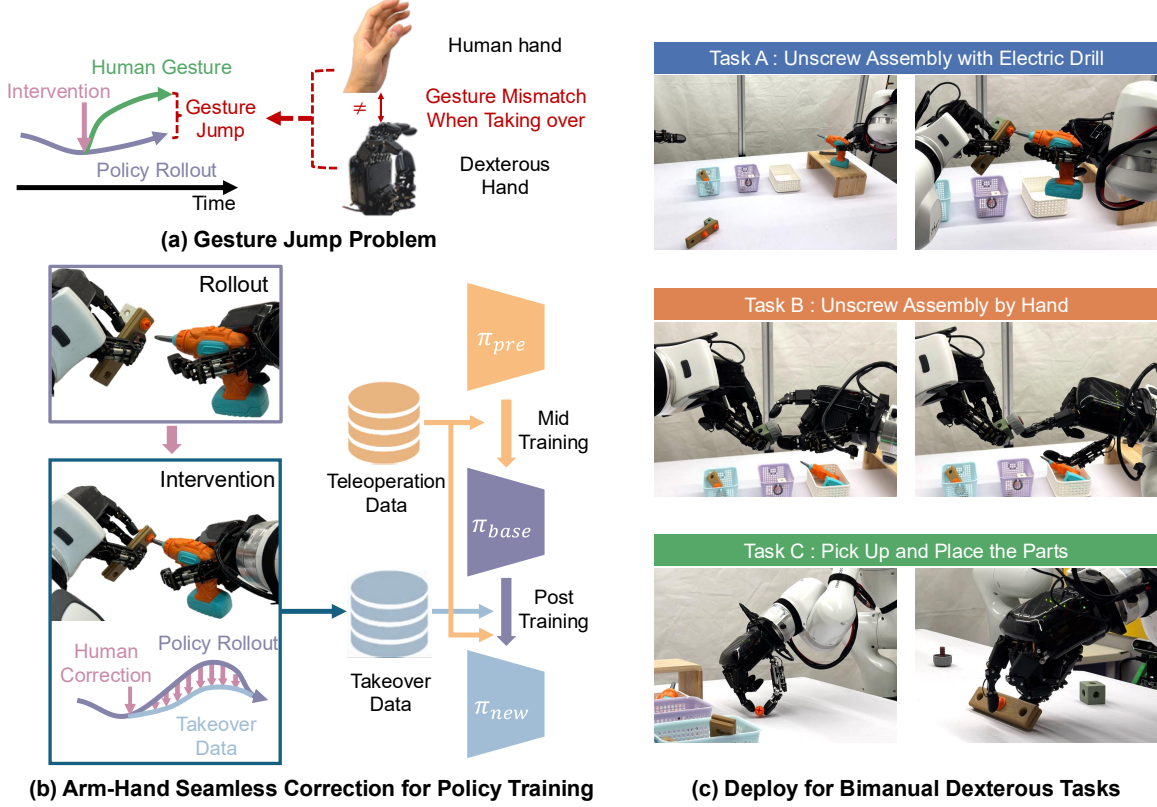


Figure 1 Overview of HandITL. (a) The gesture mismatch between human hand and robot hand at intervention moment will cause gesture jump. (b) At the intervention moment, HandITL ensures consistent command and provides smooth correction. The collected takeover data is further used to improve the policy. (c) The fine-tuned policy is deployed to execute complex, long-horizon bimanual dexterous tasks, such as unscrewing assemblies and precision part sorting.

collecting targeted data to enhance overall policy proficiency.

Despite its immense potential, deploying IIL in dexterous manipulation faces three critical bottlenecks, i.e., discontinuous “gesture jumps”, a lack of hand-arm shared autonomy, and the unverified efficacy of high-dimensional takeover data. Standard low-DoF controllers (e.g., joysticks, spacemouse) cannot express high-dimensional finger corrections. Conversely, existing high-fidelity teleoperation pipelines [23, 26] are optimized for full demonstrations rather than seamless online interventions. Consequently, sudden authority switches via traditional retargeting induce severe “gesture jumps”. As shown in Fig. 1(a), gesture jump is the command jitter caused by the gesture mismatch between human hand and robot hand when human taking over control. Because the robot’s hand may be in a stable grasp while the human’s is in a neutral pose, this abrupt command discontinuity instantly destabilizes the grasp. Such physical drops introduce unrecoverable environmental chaos, rendering subsequent task progression impossible.

To overcome these limitations, we propose HandITL, a novel seamless interventional method tailored for bimanual dexterous VLA policies, as illustrated in Fig. 1. Our core insight is the seamless fusion of human and model action intentions to enable true shared autonomy, guaranteeing minimal intervention jitter, and robust post-takeover manipulation. Specifically, for dexterous hands, we introduce an optimization-based relative retargeting method anchored at the intervention timestamp. It maps the human’s incremental motion to the robot while enforcing grasp precision, structural safety, and velocity constraints, generating smooth commands without disrupting existing contacts. Simultaneously, for the arms, we design a velocity-based shared-control interface that injects transient human wrist motions as residual twists into the VLA policy’s action trunk, enabling intuitive, and drift-free corrections.

Our system supports complex bimanual coordination, tool use, and fine-grained tasks through two modes: full takeover for critical recovery, and copilot shared control for minimal residual corrections. Extensive real-world experiments demonstrate that our method reduces command discontinuity by up to two orders of magnitude. Furthermore, fine-tuning the base VLA policy with our collected *on-policy* shared-control data substantially improves long-horizon task performance, effectively outperforming standard teleoperation fine-tuning. The main contributions of this work are summarized as follows:

- **A Seamless Interventional Method.** We identify the critical bottlenecks of “gesture jumps” and the lack of shared autonomy in dexterous IIL. In response, we propose a comprehensive method that enables online, high-DoF correction collection for long-horizon, contact-rich bimanual tasks.
- **Optimization-Based Relative Hand Retargeting.** We develop a novel retargeting algorithm that maps human relative motion to the robot. This approach ensures physical consistency during authority handovers and preserves robust manipulation capabilities post-takeover.
- **Velocity-Based Arm Shared Control.** We design a shared-control interface for bimanual arms that injects human corrective intents as transient residual twists. This mechanism allows for intuitive, drift-free corrections without disrupting the base policy’s active rollout.
- **Efficacy of On-Policy Intervention Data.** We empirically demonstrate that fine-tuning with the high-dimensional, on-policy correction data collected through our method significantly improves dexterous VLA policies, effectively mitigating compounding errors in complex tasks.

2 Related Work

2.1 VLA Models for Dexterous Manipulation

Recent VLA models have demonstrated strong generalization and instruction-following capabilities through large-scale pre-training [3, 4, 9, 10, 12, 15, 34]. However, their application to multi-fingered, high-DoF dexterous hands remains largely underexplored. Existing approaches for dexterous manipulation often rely on hierarchical systems that decouple high-level planning from low-level control [20, 30, 33]. While stable, these decoupled systems can suffer from information loss and struggle to adapt to contact-rich, long-horizon tasks. Conversely, attempts to train end-to-end VLA policies for dexterous hands are severely bottlenecked by high-dimensional action spaces, complex contact dynamics, and a critical scarcity of high-quality real-world demonstrations [2, 17, 26]. Consequently, grounding VLA capabilities into bimanual dexterous systems under real-world execution remains a formidable challenge, strongly motivating the need for post-training adaptation using on-robot feedback to mitigate compounding errors.

2.2 Interactive Imitation Learning for Policy Fine-Tuning

Deploying VLA policies on real high-DoF anthropomorphic hands inevitably faces distribution shift. As minor execution deviations accumulate during long-horizon tasks, compounding errors frequently push the robot into OOD states where the base policy fails [21, 25]. To address this, IIL—such as DAgger and its variants—has been widely adopted [1, 11, 13, 16, 21, 29]. IIL collects on-policy supervision from expert corrections during active rollouts, specifically targeting failure regions and providing explicit recovery behaviors. For simpler end-effectors like parallel-jaw grippers, this paradigm has proven highly effective in improving sample efficiency and policy robustness by aligning the training data with the policy’s actual state distribution.

However, extending this success to bimanual dexterous manipulation presents a twofold challenge. First, from a *data perspective*, it is unclear whether injecting high-dimensional corrective takeover data effectively boosts model capabilities or merely introduces detrimental interference. Second, from a *systems perspective*, exploring this data efficacy is fundamentally bottlenecked by the lack of suitable interaction interfaces. Collecting physically consistent, high-quality corrective labels for dexterous hands in real time remains a major open problem, directly motivating the need for advanced human-in-the-loop systems.

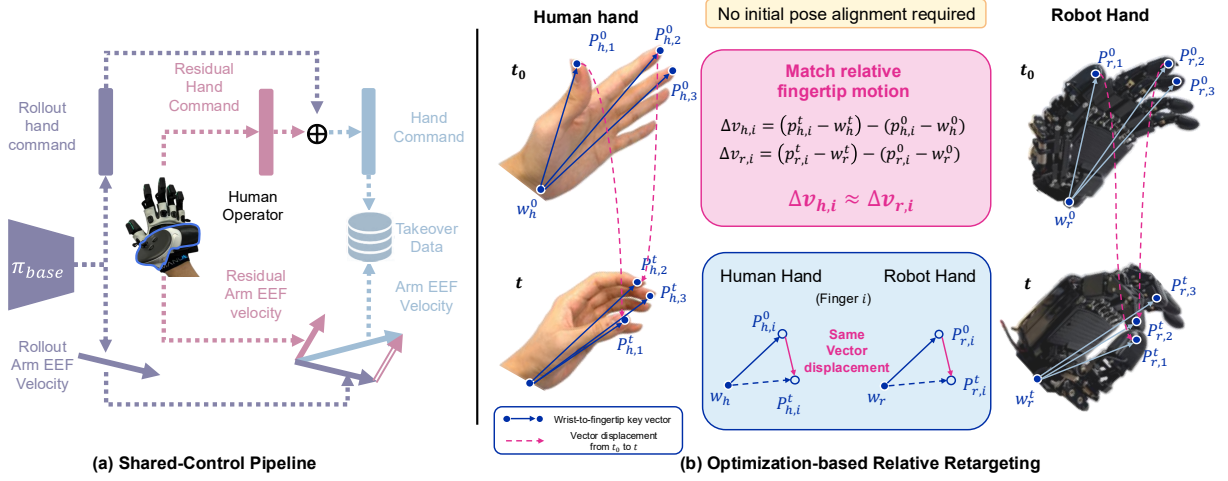


Figure 2 Architecture of the seamless interventional method. (a) The base VLA policy (π_{base}) generates autonomous rollout commands. During an intervention, the human operator injects transient residual hand commands and velocity-based arm twists. These corrective signals are seamlessly blended with the rollout commands to form the final executed actions. (b) Our method anchors at intervention moment t_0 and maps the human’s relative motion to the robot. It preserves existing contacts at the intervention moment because no robot hand motion will be caused with no human hand motion.

2.3 Interfaces for Real-Time Human Correction

Indeed, the success of any IIL paradigm heavily relies on the efficacy of its human-in-the-loop interfaces to collect these critical corrections. Standard interfaces, such as joystick [8, 22], spacemouse [5, 18], or smartphone [19], are widely used to correct low-dimensional robot trajectories. However, they are ill-suited for dexterous hands due to a severe dimensionality mismatch. High-fidelity teleoperation systems, including hand/arm exoskeletons [6, 24, 31, 32] are precise but complex and costly. Glove-based retargeting pipelines [7, 14, 23, 26], which are accessible and intuitive, are primarily optimized for collecting full demonstrations from scratch. When operators attempt to correct a rollout using its full teleoperation mode, the sudden switch in control authority abruptly disrupts the control loop. This induces discontinuous “gesture jumps” and severe command discontinuity, which instantly destabilizes contact-rich grasps. To meet the requirement for fusing human corrective intentions with the model’s ongoing rollouts, recent interfaces such as CR-Dagger [29] and RoboCopilot [28] have explored shared-control paradigms. These methods allow operators to inject targeted adjustments while preserving the base policy’s original intent. However, extending these shared-autonomy principles to high-DoF bimanual dexterous systems without triggering the aforementioned gesture jumps remains an unsolved challenge.

In contrast to these approaches, the proposed seamless interventional method utilizes data gloves and VR controllers to enable online correction without disrupting existing contacts. By employing optimization-based relative retargeting for the hands and velocity-based shared control for the arms, our system effectively eliminates command discontinuity while preserving robust manipulation capabilities post-takeover. This allows for the efficient collection of high-quality, on-policy corrective signals under both full takeover and copilot shared control modes, perfectly bridging the gap between dexterous hardware and DAgger-style fine-tuning, and verifying the efficacy of such data for long-horizon tasks.

3 Methodology

To address the challenges of post-deployment distribution shift and command discontinuity in dexterous manipulation, we propose a seamless interventional method (Fig. 2), enabling human operators to provide high-dimensional, on-policy corrections during robot execution without interrupting stable grasps.

3.1 System Overview and Interventional Interface

Our bimanual dexterous system comprises two 7-DoF Franka FR3 robotic arms, each equipped with a 21-DoF ByteDexter V2 hand [26], forming a high-dimensional action space suitable for complex manipulation tasks. To facilitate seamless online intervention, we construct a multimodal human-in-the-loop interface. The operator’s wrist poses (6-DoF) are tracked using a Meta Quest 3 headset and controllers, while fine-grained finger movements are captured via Manus Quantum Metagloves. The Quest controllers are rigidly mounted to the back of the Manus gloves using custom-designed fixtures. We integrate a dual-pedal controller that allows the operator to seamlessly switch between autonomous rollout and hands-free human intervention modes. For perception, the workspace is covered by three global RGB-D cameras providing complementary views: two egocentric views (top-down and bottom-up) and one third-person view.

3.2 Problem Formulation and Intervention Modes

We formulate bimanual dexterous manipulation as a sequential decision-making process parameterized by a VLA policy π_θ . At each time step t , the policy receives an observation o_t (comprising RGB-D images and proprioception) and predicts an action sequence $\mathbf{A}_t^\pi$ with horizon H .

During standard autonomous execution, the robot executes the first action of the predicted sequence $\mathbf{A}_t^\pi$. In our human-in-the-loop setting, the operator can inject a corrective action $\mathbf{a}_t^h$ during the rollout. The final executed command $\mathbf{a}_t^{exec}$ is a blended output: $\mathbf{a}_t^{exec} = \alpha \mathbf{a}_t^\pi + \beta \mathbf{a}_t^h$ where $\alpha \in [0, 1]$ denotes the policy weight and $\beta \in [0, 1]$ denotes the human authority weight. The details of fusing arm and hand command are shown in Fig. 2(a). Our method supports two primary intervention modes:

- **Full Takeover:** $\alpha \approx 0, \beta = 1$. The robot is primarily controlled by the human, which is crucial for recovering from severe OOD states.
- **Copilot Shared Control:** The operator injects minimal corrective signals while preserving the base policy’s intent. The system allows decoupled authority weights for the arms (β_a) and hands ($\beta_h = 1 - \alpha_h$).

Through iterative fine-tuning, these rollout interventions are aggregated into a correction dataset $\mathcal{D}_{correct} = \{(o_t, \mathbf{a}_t^{exec})\}_{t=1}^T$. Because our method ensures that $\mathbf{a}_t^{exec}$ is physically consistent and smooth, this on-policy data is highly effective for updating the base policy π_{base} .

3.3 Optimization-Based Relative Hand Retargeting

Traditional absolute retargeting maps human poses directly to the robot, causing sudden “gesture jumps” during interventions that destabilize contact-rich grasps. To eliminate this, we propose an **optimization-based relative retargeting** algorithm. At each control step t , we solve for a target hand configuration q_t that tracks the human’s relative **keyvectors** while enforcing safety constraints. To ensure robustness against tracking outliers, we employ the Huber loss $\mathcal{H}_\delta(\cdot)$. The optimization objective $\mathcal{J}(q_t)$ is formulated as:

$$\begin{aligned} \min_{q_t} \mathcal{J}(q_t) = & \underbrace{\sum_{i=1}^5 \mathcal{H}_\delta\left(\beta_i(d_i)(\Delta \mathbf{v}_i^{\text{rob}}(q_t) - \Delta \hat{\mathbf{v}}_i^{\text{hum}})\right)}_{\mathcal{L}_{\text{shape}} \text{ (Global Shaping)}} + \underbrace{\sum_{j \in \mathcal{F}_{\text{opp}}} \mathcal{H}_\delta\left(\omega_j(d_j)(\mathbf{u}_j^{\text{rob}}(q_t) - \alpha_j(d_j)\hat{\mathbf{u}}_j^{\text{tgt}})\right)}_{\mathcal{L}_{\text{grasp}} \text{ (Precision Grasping)}} \\ & + \underbrace{\sum_{(m,n) \in \mathcal{C}} \gamma \max\left(0, \frac{d_{\text{safe}}}{D_{mn}(q_t)} - 1\right)}_{\mathcal{L}_{\text{safe}} \text{ (Structural Safety)}} + \underbrace{\mathcal{H}_\delta\left(\lambda_{\text{reg}}(q_t - q_{t-1})\right)}_{\mathcal{L}_{\text{reg}} \text{ (Temporal Regularization)}} , \end{aligned} \quad (1)$$

where superscripts rob and hum denote robot and human states, and $\Delta(\cdot) = (\cdot)_t - (\cdot)_{t_0}$ represents the relative motion since the intervention timestamp t_0 . The loss components are defined as follows:

- **Global Shaping ($\mathcal{L}_{\text{shape}}$):** Prioritizes overall posture via **wrist-to-fingertip keyvectors** $\mathbf{v}_i$. Here, $\Delta \mathbf{v}_i^{\text{rob}}(q_t)$ and $\Delta \hat{\mathbf{v}}_i^{\text{hum}}$ are the relative changes of these vectors on the robot and human hands. The adaptive weight

$\beta_i(d_i)$ decays via a sigmoid as the absolute human thumb-to-finger distance d_i decreases, dominating when the hand is open.

- **Precision Grasping ($\mathcal{L}_{\text{grasp}}$):** Captures pinch grasping via **thumb-to-fingertip keyvectors** $\mathbf{u}_j$. The robot’s absolute keyvector $\mathbf{u}_j^{\text{rob}}(q_t)$ tracks the nominal target $\hat{\mathbf{u}}_j^{\text{tgt}} = \mathbf{u}_j^{\text{rob}}(t_0) + \Delta \hat{\mathbf{u}}_j^{\text{hum}}$. We introduce a pinch activation factor $\alpha_j(d_j) \in [0, 1]$. When d_j falls below a threshold, $\alpha_j(d_j) \rightarrow 0$, explicitly driving $\mathbf{u}_j^{\text{rob}}$ to $\mathbf{0}$ (i.e., firm contact). Concurrently, the weight $\omega_j(d_j) \propto (1 - \alpha_j)$ sharply increases, ensuring this term dominates during closed grasps.
- **Structural Safety ($\mathcal{L}_{\text{safe}}$):** A repulsive barrier preventing self-collisions. $D_{mn}(q_t)$ is the absolute physical distance between adjacent PIP/DIP joint pairs $(m, n) \in \mathcal{C}$. The penalty γ activates only when D_{mn} falls below the safety margin d_{safe} .
- **Temporal Regularization ($\mathcal{L}_{\text{reg}}$):** Penalizes the joint displacement $(q_t - q_{t-1})$, scaled by λ_{reg} , to prevent high-frequency jitter.

3.4 Velocity-Based Shared Arm Control

Since VR controllers are rigidly mounted on the gloves, operators lack access to physical joysticks. To enable intuitive corrections, we design a **velocity-based shared-control** interface. Human wrist motions are translated into residual twists, smoothed via Exponential Moving Average (EMA), and injected into the policy trunk without persistent drift.

At step t , the VLA outputs the target pose and feedforward twist $(\mathbf{p}_t^\pi, \mathbf{R}_t^\pi, \dot{\mathbf{p}}_t^\pi, \boldsymbol{\omega}_t^\pi)$. Simultaneously, the residual human twist $(\dot{\mathbf{p}}_t^h, \boldsymbol{\omega}_t^h)$ is estimated over a window of $k = 2$ VR ticks ($\Delta T = 0.04$ s) to balance stability and latency:

$$\dot{\mathbf{p}}_t^h = \text{EMA} \left(\frac{\mathbf{p}_t^{vr} - \mathbf{p}_{t-k}^{vr}}{\Delta T} \right), \quad (2)$$

$$\boldsymbol{\omega}_t^h = \text{EMA} \left(\frac{\log((\mathbf{R}_{t-k}^{vr})^{-1} \mathbf{R}_t^{vr})}{\Delta T} \right). \quad (3)$$

These twists are integrated into incremental pose offsets $(\Delta \mathbf{p}_t^h, \Delta \mathbf{R}_t^h)$ to compute the final corrective target pose:

$$\mathbf{p}_t^{\text{tgt}} = \mathbf{p}_t^\pi + \Delta \mathbf{p}_t^h, \quad \mathbf{R}_t^{\text{tgt}} = \mathbf{R}_t^\pi \Delta \mathbf{R}_t^h. \quad (4)$$

A task-space PD tracker, augmented with policy feedforward terms, then computes the commanded spatial velocity.

Crucially, deriving residuals from relative motion ensures that when the operator is still, $(\dot{\mathbf{p}}_t^h, \boldsymbol{\omega}_t^h) \rightarrow \mathbf{0}$. The EMA filter smoothly decays these residuals, eliminating persistent offsets without requiring the operator to manually return to a neutral “center” position.

4 Experiments

We aim to answer the following research questions through experiments:

- **Q1:** Does our optimization-based relative retargeting method effectively reduce *command discontinuity* (i.e., gesture jumps) at the moment of takeover?
- **Q2:** Does our human-in-the-loop interface preserve sufficient high-DoF manipulation capability after the takeover occurs?
- **Q3:** Can the on-policy correction data collected via our method—especially under *copilot shared control*—improve VLA policy learning when real-world data is limited?
- **Q4:** Can the overall method support challenging, contact-rich, long-horizon bimanual dexterous tasks?

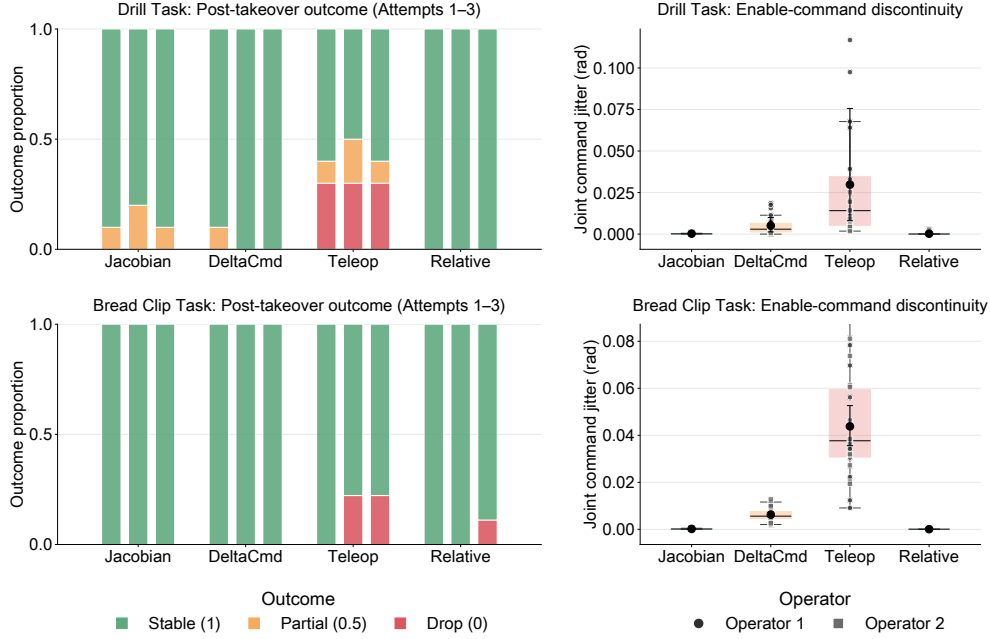


Figure 3 Takeover command discontinuity on the Drill (top) and Bread Clip (bottom) tasks. Left (Grasp Stability): Outcomes across three consecutive takeover attempts (10 trials each). Green, orange, and red denote stable grasps, partial failures (trigger loosening), and tool drops, respectively. **Right (Command Change):** Attempt-level distribution of the mean command change at intervention. Scatter points show individual takeovers from two operators, with 95% CI error bars.

4.1 Experiments Overview

Our experiments consist of three parts: (1) takeover command discontinuity analysis, which measures the abrupt change in dexterous hand commands at the handover moment; (2) post-takeover manipulation evaluation, which tests whether each method preserves sufficient dexterous manipulation capability after takeover; and (3) long-horizon policy evaluation, which examines whether the collected human-correction data improves the performance of a VLA policy on challenging bimanual dexterous manipulation tasks.

For the first two parts, we compare our optimization-based relative retargeting method with three alternative takeover strategies.

Jacobian-based mapping: This method computes joint increments from fingertip displacements using the Jacobian at each discrete time step[23].

Delta-command retargeting: This method uses the teleoperation retargeting pipeline as its backend[7, 27], but applies the command difference. Suppose takeover happens at time t and the current robot hand state is q_0 . Let the retargeted glove commands at times t and $t + 1$ be q_t and q_{t+1} , respectively. Then the executed command at time $t + 1$ is computed as $q_{t+1}^{exec} = q_0 + (q_{t+1} - q_t)$

Direct teleoperation switching: This method directly switches control authority to the teleoperation retargeting pipeline at takeover[27].

In the third part, we deploy the base policy on real robots, collect takeover data when the operator considers intervention necessary, and then use these data for post-training. We evaluate the resulting policies on three long-horizon dexterous manipulation tasks.

Table 1 Post-takeover manipulation capability on the **Pick Up and Place the Parts** and **Pick Up the Drill** tasks. **Interv.** (Interventions) and **Retries** are reported as the total count across all 10 trials (5 per operator). **Time (Op1/Op2)** denotes the separate mean completion times for Operator 1 and Operator 2 to demonstrate cross-operator robustness. **Success** rate indicates whether the drill trigger was successfully actuated.

Method	Task 1: Pick Up and Place the Parts				Task 2: Pick Up the Drill			
	Time (s) ↓	Time (Op1/Op2) ↓	Interv. ↓	Retries ↓	Time (s) ↓	Time (Op1/Op2) ↓	Interv. ↓	Success ↑
Jacobian	68.0 ± 10.8	66.5 / 69.5	31/10	14/10	19.4 ± 5.6	20.3 / 18.5	14/10	7/10
Delta Cmd	56.7 ± 14.5	58.0 / 55.5	23/10	9/10	17.3 ± 5.4	13.1 / 21.5	11/10	8/10
Teleop	52.9 ± 14.2	43.9 / 61.9	21/10	8/10	15.5 ± 5.7	12.2 / 18.9	12/10	10/10
Relative (Ours)	42.8 ± 5.0	40.0 / 45.6	17/10	1/10	14.4 ± 4.7	11.0 / 17.9	11/10	8/10

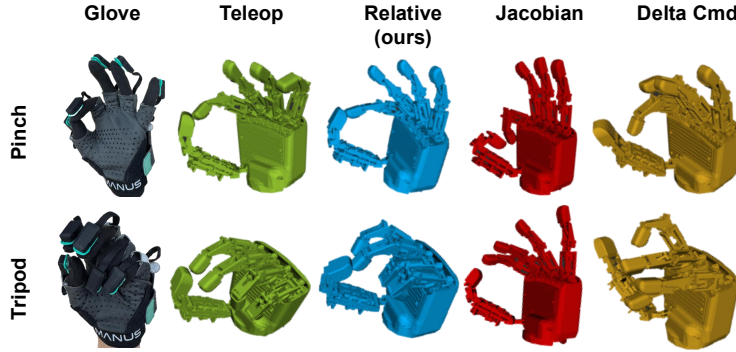


Figure 4 Grasping Postures and Failure Modes. Optimization-based methods (Teleop and our Relative approach) maintain precise thumb-to-finger opposition and avoid collisions. In contrast, differential methods (Jacobian and Delta Cmd) struggle with pinch grasping and suffer from accumulated drift during fast motions, leading to awkward postures and frequent grasp failures.

4.2 Takeover Command Discontinuity Analysis

We evaluate takeover command discontinuity on the Drill and Bread Clip tasks. Operators first establish a stable teleoperation on the tool, switch to rollout mode, and then toggle the takeover mode three times per trial.

Across 30 attempts per task-method pair, direct teleoperation switching exhibited a high failure rate, frequently dropping the tool or releasing the drill trigger due to abrupt gesture jumps. In contrast, our relative retargeting method seamlessly preserved grasp stability. Notably, the single recorded failure for our method on the Bread Clip task resulted from an operator manipulation error after a successful handover, rather than a command jump at the exact intervention timestamp.

As shown in Fig. 3, our method drastically reduces quantitative command mutation at intervention. On the Bread Clip task, which requires an open hand posture that maximizes human-robot state mismatch, direct switching introduces massive command spikes (mean $\approx 4.38 \times 10^{-2}$). Our method suppresses this to $\approx 6.8 \times 10^{-5}$, reducing the discontinuity by **99.8%** and outperforming the DeltaCmd baseline ($\approx 6.23 \times 10^{-3}$) by nearly two orders of magnitude. Conversely, the Drill task involves a closed power grasp, naturally restricting the maximum pose mismatch and yielding numerically smaller jumps for direct switching ($\approx 2.75 \times 10^{-2}$). However, even this seemingly minor mutation is catastrophic, instantly causing the index finger to slip off the trigger. Our method resolves this by achieving a near-zero mean command change ($\approx 2.65 \times 10^{-4}$), comparable to the Jacobian baseline but without its inherent singularity issues, ensuring stable trigger actuation and functional tool use.



Figure 5 Execution sequences of the three long-horizon bimanual dexterous manipulation tasks. Top: *Unscrew Assembly With Electric Drill*, requiring tool use and bimanual coordination. Middle: *Unscrew Assembly By Hand*, demanding fine-grained in-hand manipulation. Bottom: *Pick Up and Place the Parts*, involving sequential precision grasping of diverse small objects.

4.3 Post-Takeover Manipulation Capability

We evaluate pure manipulation capability during takeover on the **Pick Up and Place the Parts** and **Pick Up the Drill** tasks in Fig. 5. Operators initiated both tasks directly in an active takeover state. If an operator reached their physical workspace limit, they could temporarily disengage, reset their hand posture, and re-engage (recorded as *Interventions*).

As shown in Table. 1, on the Pick Up and Place the Parts task, our method achieves the fastest mean time (42.8s), improving by 19.1%, and reduces grasp failures (retries) by 87.5% compared to Teleop. Furthermore, our method exhibits minimal cross-operator variance (a 5.6s gap vs. 18.0s for Teleop), proving its high intuitiveness. As illustrated in Fig. 4, this performance gap stems from the underlying kinematics. Both our Relative method and Teleop utilize optimization algorithms, ensuring accurate thumb-to-finger opposition and collision avoidance. Conversely, the Jacobian and Delta Cmd baselines rely on previous-step differential calculations. They struggle with precise pinching and easily accumulate drift during fast motions, resulting in awkward grasping postures, frequent part drops, and significantly longer completion times (68.0s and 56.7s, respectively). On the Pick Up the Drill task, our method again demonstrates better overall efficiency. It records the lowest mean completion time (14.4s), the fewest workspace interventions (11/10) and maintains better cross-operator time consistency (11.0s / 17.9s) compared to the differential baselines. Because our relative retargeting employs a different optimization formulation than the teleoperation, the mapping of fine finger articulation differs slightly. Consequently, without tactile feedback, operators require a brief period of adaptation and postural adjustment to command, which slightly affect success rate of triggering the button. Overall, our method preserves highly efficient and robust dexterous manipulation capabilities.

4.4 Long-Horizon Policy Evaluation

We collect a 20-hour real-world teleoperation dataset, evenly distributed across three long-horizon tasks. Starting from a pre-trained GR-Dexter policy [26], we further fine-tune the model on this teleoperation dataset

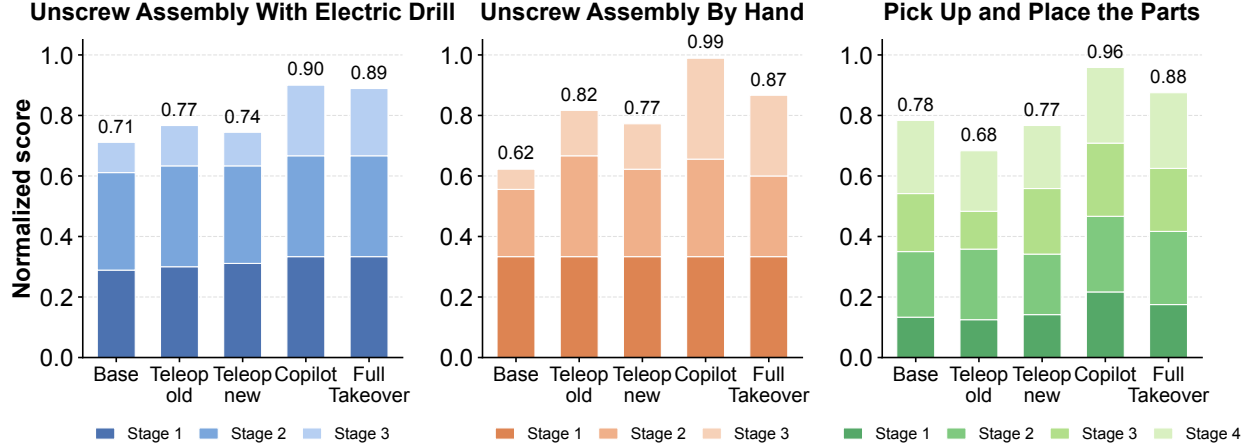


Figure 6 Average normalized sub-goal completion scores across three long-horizon tasks. Intervention-trained policies (Copilot, Full Takeover) outperform pure teleoperation baselines, indicating improved robustness against compounding errors.

to obtain our *base policy*. We then deploy the base policy on real robots and collect takeover data during rollouts whenever the operator judges intervention to be necessary. We collect correction data under both full-takeover and copilot shared-control modes in an HG-Dagger-style takeover process. For post-training, we compare the following policies:

- **Base:** the policy fine-tuned only on the 20-hour teleoperation dataset (base dataset);
- **Teleop_old:** continuing to fine-tune the base policy using the base dataset only;
- **Teleop_new:** fine-tuning the base policy using a 0.5:1 mixture of 1-hour new teleoperation data and base dataset;
- **Copilot:** fine-tuning the base policy using a 0.5:1 mixture of copilot takeover data and base dataset;
- **Full Takeover:** fine-tuning the base policy using a 0.5:1 mixture of full-takeover data and base dataset.

For a fair comparison, all additional post-training datasets, including new teleoperation, copilot takeover, and full-takeover data, are 1 hour in duration, and all post-training runs use identical training schedules and hyperparameters.

We evaluate all policies on three long-horizon tasks requiring high-precision bimanual coordination (Fig. 5). Each task is decomposed into several semantic sub-stages. We report the average Sub-goal Completion Score over 10 evaluation rollouts for each model.

As shown in Fig. 6, comparing the five policies reveals three critical insights regarding long-horizon performance:

1) Limited Improvements from Pure Teleoperation: As shown in Fig. 6, simply increasing pure teleoperation data brings limited and inconsistent gains. Both **Teleop_old** and **Teleop_new** show only marginal improvements, and their effects vary across tasks. This suggests that additional off-policy demonstrations do not sufficiently address the rollout states where compounding errors occur, especially in late phases that require precise contact-rich manipulation.

2) Effectiveness of Intervention Data: In contrast, policies fine-tuned with intervention data achieve higher average normalized completion scores. Since **Copilot** and **Full Takeover** data are collected when the deployed policy actually requires human correction, they provide targeted supervision for recovering from out-of-distribution states. This makes intervention data more effective than standard teleoperation demonstrations for improving long-horizon robustness.

3) Copilot vs. Full Takeover: Among the two intervention strategies, **Copilot** generally yields the strongest

overall performance. Unlike **Full Takeover**, which may introduce larger action deviations during authority switching, **Copilot** preserves the base policy’s ongoing behavior while injecting local corrections. The resulting data stays closer to the policy’s rollout distribution, leading to more stable downstream improvements.

5 Conclusion and Limitations

Conclusion. In this work, we presented a seamless interventional method tailored for bimanual high-DoF dexterous VLA policies to address brittle deployment and compounding errors. By identifying “gesture jumps” and the lack of true shared autonomy as critical bottlenecks in existing IIL pipelines, we introduced an optimization-based relative hand retargeting method and a velocity-based shared arm control interface. Extensive real-world experiments demonstrated that our method reduces command discontinuity by up to two orders of magnitude, preserving robust manipulation capabilities post-takeover. Furthermore, we verified the efficacy of high-dimensional takeover data: the on-policy corrections collected via our Copilot shared control act as a highly efficient patch for OOD states. By blending this data, we significantly improved the sub-goal completion rates of complex, long-horizon tasks over standard teleoperation and full takeover baselines.

Limitations and Future Work. Despite these advances, two main limitations remain. First, our current post-training strategy relies on simple supervised fine-tuning. Since human intervention data inherently contains sub-optimal recovery trajectories or panic-induced noise, how to more efficiently utilize this data remains an open question. Future work could explore automated data filtering or preference learning methods to extract more robust learning signals. Second, the policy’s capability in extremely high-precision manipulation—such as millimeter-level drill bit alignment—remains limited. This bottleneck is primarily due to visual occlusions and the spatial resolution limits of vision-only VLA architectures, highlighting the need to integrate multi-modal sensing in future iterations.

References

- [1] Yunus Bicer, Ali Alizadeh, Nazim Kemal Ure, Ahmetcan Erdogan, and Orkun Kizilirmak. Sample efficient interactive end-to-end deep learning for self-driving cars with selective multi-class safe dataset aggregation. In 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pages 2629–2634. IEEE, 2019.
- [2] Johan Bjorck, Fernando Castañeda, Nikita Cherniadev, Xingye Da, Runyu Ding, Linxi Fan, Yu Fang, Dieter Fox, Fengyuan Hu, Spencer Huang, et al. Gr00t n1: An open foundation model for generalist humanoid robots. arXiv preprint arXiv:2503.14734, 2025.
- [3] Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, et al. π_0 : A vision-language-action flow model for general robot control. arXiv preprint arXiv:2410.24164, 2024.
- [4] Chilam Cheang, Sijin Chen, Zhongren Cui, Yingdong Hu, Liquan Huang, Tao Kong, Hang Li, Yifeng Li, Yuxiao Liu, Xiao Ma, et al. Gr-3 technical report. arXiv preprint arXiv:2507.15493, 2025.
- [5] Yuhui Chen, Shuai Tian, Shugao Liu, Yingting Zhou, Haoran Li, and Dongbin Zhao. Conrft: A reinforced fine-tuning method for vla models via consistency policy. arXiv preprint arXiv:2502.05450, 2025.
- [6] Moein Forouhar, Hamid Sadeghian, Daniel Perez Suay, Abdeldjalil Naciri, and Sami Haddadin. A tactile lightweight exoskeleton for teleoperation: Design and control performance. In 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pages 178–183. IEEE, 2024.
- [7] Ankur Handa, Karl Van Wyk, Wei Yang, Jacky Liang, Yu-Wei Chao, Qian Wan, Stan Birchfield, Nathan Ratliff, and Dieter Fox. Dexpivot: Vision-based teleoperation of dexterous robotic hand-arm system. In 2020 IEEE International Conference on Robotics and Automation (ICRA), pages 9164–9170. IEEE, 2020.
- [8] Zheyuan Hu, Robyn Wu, Naveen Enock, Jasmine Li, Riya Kadakia, Zackory Erickson, and Aviral Kumar. Rac: Robot learning for long-horizon tasks by scaling recovery and correction. arXiv preprint arXiv:2509.07953, 2025.
- [9] Physical Intelligence, Ali Amin, Raichelle Aniceto, Ashwin Balakrishna, Kevin Black, Ken Conley, Grace Connors, James Darpinian, Karan Dhabalia, Jared DiCarlo, et al. $\pi_{0,6}^*$: a vla that learns from experience. arXiv preprint arXiv:2511.14759, 2025.
- [10] Physical Intelligence, Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, et al. $\pi_{0,5}$: a vision-language-action model with open-world generalization. arXiv preprint arXiv:2504.16054, 2025.
- [11] Michael Kelly, Chelsea Sidrane, Katherine Driggs-Campbell, and Mykel J Kochenderfer. Hg-dagger: Interactive imitation learning with human experts. In 2019 International Conference on Robotics and Automation (ICRA), pages 8077–8083. IEEE, 2019.
- [12] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, et al. Openvla: An open-source vision-language-action model. arXiv preprint arXiv:2406.09246, 2024.
- [13] Sung-Wook Lee, Xuhui Kang, and Yen-Ling Kuo. Diff-dagger: Uncertainty estimation with diffusion policy for robotic manipulation. In 2025 IEEE International Conference on Robotics and Automation (ICRA), pages 4845–4852. IEEE, 2025.
- [14] Shuang Li, Norman Hendrich, Hongzhuo Liang, Philipp Ruppel, Changshui Zhang, and Jianwei Zhang. A dexterous hand-arm teleoperation system based on hand pose estimation and active vision. IEEE Transactions on Cybernetics, 54(3):1417–1428, 2022.
- [15] Yunfei Li, Xiao Ma, Jiafeng Xu, Yu Cui, Zhongren Cui, Zhigang Han, Liquan Huang, Tao Kong, Yuxiao Liu, Hao Niu, et al. Gr-rl: Going dexterous and precise for long-horizon robotic manipulation. arXiv preprint arXiv:2512.01801, 2025.
- [16] Deqing Liu, Yinfeng Gao, Deheng Qian, Qichao Zhang, Xiaoqing Ye, Junyu Han, Yupeng Zheng, Xueyi Liu, Zhongpu Xia, Dawei Ding, et al. Takead: Preference-based post-optimization for end-to-end autonomous driving with expert takeover data. IEEE Robotics and Automation Letters, 11(2):1738–1745, 2025.

- [17] Hao Luo, Yicheng Feng, Wanpeng Zhang, Sipeng Zheng, Ye Wang, Haoqi Yuan, Jiazheng Liu, Chaoyi Xu, Qin Jin, and Zongqing Lu. Being-h0: vision-language-action pretraining from large-scale human videos. [arXiv preprint arXiv:2507.15597](#), 2025.
- [18] Jianlan Luo, Charles Xu, Jeffrey Wu, and Sergey Levine. Precise and dexterous robotic manipulation via human-in-the-loop reinforcement learning. *Science Robotics*, 10(105):eads5033, 2025.
- [19] Ajay Mandlekar, Danfei Xu, Roberto Martín-Martín, Yuke Zhu, Li Fei-Fei, and Silvio Savarese. Human-in-the-loop imitation learning using remote teleoperation. [arXiv preprint arXiv:2012.06733](#), 2020.
- [20] Xiaofeng Mao, Gabriele Giudici, Claudio Coppola, Kaspar Althoefer, Ildar Farkhatdinov, Zhibin Li, and Lorenzo Jamone. Dexskills: Skill segmentation using haptic data for learning autonomous long-horizon robotic manipulation tasks. In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 5104–5111. IEEE, 2024.
- [21] Stéphane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the fourteenth international conference on artificial intelligence and statistics*, pages 627–635. JMLR Workshop and Conference Proceedings, 2011.
- [22] Jonathan Spencer, Sanjiban Choudhury, Matthew Barnes, Matthew Schmittle, Mung Chiang, Peter Ramadge, and Siddhartha Srinivasa. Learning from interventions. In *Robotics: Science and Systems (RSS)*, volume 1, page 2, 2020.
- [23] Chen Wang, Haochen Shi, Weizhuo Wang, Ruohan Zhang, Li Fei-Fei, and C Karen Liu. Dexcap: Scalable and portable mocap data collection system for dexterous manipulation. [arXiv preprint arXiv:2403.07788](#), 2024.
- [24] Dehao Wei and Huazhe Xu. A wearable robotic hand for hand-over-hand imitation learning. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 18113–18119. IEEE, 2024.
- [25] Edgar Welte and Rania Rayyes. Interactive imitation learning for dexterous robotic manipulation: challenges and perspectives—a survey. *Frontiers in Robotics and AI*, 12:1682437, 2025.
- [26] Ruoshi Wen, Guangzeng Chen, Zhongren Cui, Min Du, Yang Gou, Zhigang Han, Liqun Huang, Mingyu Lei, Yunfei Li, Zhuohang Li, et al. Gr-dexter technical report. [arXiv preprint arXiv:2512.24210](#), 2025.
- [27] Ruoshi Wen, Jiajun Zhang, Guangzeng Chen, Zhongren Cui, Min Du, Yang Gou, Zhigang Han, Junkai Hu, Liqun Huang, Hao Niu, et al. Dexterous teleoperation of 20-dof bytedexter hand via human motion retargeting. [arXiv preprint arXiv:2507.03227](#), 2025.
- [28] Philipp Wu, Yide Shentu, Qiayuan Liao, Ding Jin, Menglong Guo, Koushil Sreenath, Xingyu Lin, and Pieter Abbeel. Robocopilot: Human-in-the-loop interactive imitation learning for robot manipulation. [arXiv preprint arXiv:2503.07771](#), 2025.
- [29] Xiaomeng Xu, Yifan Hou, Zeyi Liu, and Shuran Song. Compliant residual dagger: Improving real-world contact-rich manipulation with human corrections. [arXiv preprint arXiv:2506.16685](#), 2025.
- [30] Haoqi Yuan, Yu Bai, Yuhui Fu, Bohan Zhou, Yicheng Feng, Xinrun Xu, Yi Zhan, Börje F Karlsson, and Zongqing Lu. Being-0: A humanoid robotic agent with vision-language models and modular skills. [arXiv preprint arXiv:2503.12533](#), 2025.
- [31] Han Zhang, Songbo Hu, Zhecheng Yuan, and Huazhe Xu. Doglove: Dexterous manipulation with a low-cost open-source haptic force feedback glove. [arXiv preprint arXiv:2502.07730](#), 2025.
- [32] Rui Zhong, Chuang Cheng, Junpeng Xu, Yantong Wei, Ce Guo, Daoxun Zhang, Wei Dai, and Huimin Lu. Nuexo: A wearable exoskeleton covering all upper limb rom for outdoor data collection and teleoperation of humanoid robots. In *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 12026–12033. IEEE, 2025.
- [33] Yifan Zhong, Xuchuan Huang, Ruochong Li, Ceyao Zhang, Zhang Chen, Tianrui Guan, Fanlian Zeng, Ka Nam Lui, Yuyao Ye, Yitao Liang, et al. Dexgraspv1a: A vision-language-action framework towards general dexterous grasping. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 18836–18844, 2026.
- [34] Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, Jialin Wu, Paul Wohlhart, Stefan Welker, Ayzaan Wahid, et al. Rt-2: Vision-language-action models transfer web knowledge to robotic control. In *Conference on Robot Learning*, pages 2165–2183. PMLR, 2023.